

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester VI Examination, 2022-23

Paper No. MTB-603: Numerical Analysis

Time: 3 Hours

Full Marks: 50

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks. Use of calculator is allowed. All symbols have their usual meanings.*

Question 1.

- (a) A can is in the form of a right circular cylinder of radius 15 cm and length 40 cm, surmounted by a hemisphere. If the scale used in taking measurements is short by 0.01 cm per cm, find the percentage change in the volume of the can. [2]
- (b) If $r = 3h(n^2 + 2)$, find the percentage error in r at $h = 1$, given that the percentage error in h is 0.5%. [2]
- (c) Prove or disprove: (i) $\Delta = (1 + \Delta)^{1/2}$, (ii) $abla\Delta = ab\nabla - \Delta$. [2]
- (d) If $\Delta^2(\cos 2x) = A \cos 2(x + h)$, where h is the interval of differencing, find the value of A . [2]
- (e) Apply Simpson's rule to find $\int_0^{2.5} f(x) dx$, given:

x	0	0.5
1	2.5	
$f(x)$	0	0.25
1	2.25	

[2]

Question 2.

- (a) Apply the Secant method to find the numerically largest root, correct to two decimal places, of $x^3 - 5x + 3 = 0$. [5]
- (b) Determine p, q, r so that the iterative method $x_{n+1} = px_n + \frac{q}{x_n} + \frac{r}{x_n^2}$ has as high an order of convergence as possible for computing the cube root of a . [5]

Question 3.

- (a) Use three iterations of the Birge-Vieta method (calculations to five decimal places) to find a real root of $x^3 - x + 1 = 0$, starting with initial approximation $x_0 = -1.5$. [5]

- (b) Solve the system $3x + 5y + z = 7$, $x + y + 3z = 3$, $4x + y + 2z = 4$ using the Gauss-Jacobi method (correct to three decimal places) starting with $[0, 0, 0]^T$. [5]
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Question 4.

- (a) Apply Givens' method to find all eigenvalues of:

$$A = \begin{pmatrix} -2 & -2 & 6 \\ -2 & 3 & 4 \\ 6 & 4 & -1 \end{pmatrix}$$

[5]

- (b) Using finite differences, find the sum of cubes of the first m natural numbers. [5]
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Question 5.

- (a) Apply Lagrange's formula inversely to find the root of $f(x) = 0$, given $f(30) = -30$, $f(34) = -13$, $f(38) = 3$, $f(42) = 18$. [5]
- (b) Using Newton's forward interpolation formula, estimate the number of workers earning wages between Rs.10 and Rs.15 per hour from the given frequency table. [5]
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Question 6.

- (a) Find the first and second derivatives of $y(x)$ at $x = 1.5$ from the given tabular data. [5]
- (b) Obtain the error term in the Trapezoidal rule of numerical integration. [5]
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Question 7.

- (a) A reservoir discharges water through sluices at depth h below the water surface with surface area A . The rate of fall of the surface is given by $\frac{dh}{dt} = -\frac{48\sqrt{h}}{A}$. Estimate the time taken for the water level to fall from 14 m to 10 m above the sluices using Simpson's rule. [5]
- (b) Use the fourth-order Runge-Kutta method to evaluate $y(1.2)$ and $y(1.4)$, given $\frac{dy}{dx} = \frac{y}{x} + \frac{1}{x}$, $y(1) = 0$. [5]
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